Safeguarding Life · Public Health

JLU-SPH

2026 "Synthetic Biology Students' Ecosystem" Future Career Seminar Planning Proposal

Protecting life with the heart of science

Responding to public health through collaborative strength





JILIN UNIVERSITY · SCHOOL OF PUBLIC HEALTH

Ⅰ、 Basic Information of the Event

- Event Name: "Synthetic Biology Ecosystem" Future Career Symposium
- Event Theme: Centered on the development of a "respiratory tract detection invention", exploring the limitless possibilities of the synthetic biology full-industry-chain ecosystem
- Event Type: ☑ Science Education ☐ Public Welfare ☐ Cultural Promotion ☐ Community Service ☐ Other:
- Organizer: JLU-SPH Team

Ⅱ、 Purpose of the Event

1. Core Purpose: To demonstrate to secondary school students that synthetic biology is more than just laboratory research—it is a vast ecosystem involving diverse roles, including ethics, law, business, and more.
2. Specific Objectives:

○ Knowledge/Cognitive Level: Using the development and implementation of a "pathogen detection product" as an example, enabling students to recognize the necessity of multi-faceted professionals such as bioethicists and biosafety legal specialists in technological innovation.

○ Behavioral/Skill Level: Students should be able to position their potential future roles within the "Career Ecosystem Tree" based on their individual strengths.

○ Emotional/Attitudinal Level: To dispel the misconception that participation in synthetic biology is exclusive to scientists, and to inspire students from varied academic backgrounds to engage enthusiastically.

3. Expected Outcomes:

○ Number of Participants: 50–80 high school students

○ Satisfaction Target: Over 90%

○ Jointly create and produce a comprehensive “Career Ecosystem Tree ” chart.

Ⅲ、 Target Audience

- Core Audience: Secondary school students
- Audience Characteristics: Students are in a period of career planning confusion, interested in biotechnology but possibly deterred due to lack of laboratory skills,and eager for inspiration from diverse perspectives

Ⅳ、 Date, Time and Venue

- Event Date: May 11th, 2026
- Total Duration: Approximately 40 minutes
- Venue: Multimedia classroom of partner high school

Ⅴ、 Event Content and Process

[Overview of Event Flow]

Phase	Time	Session Name	Specific Content	Responsible Personnel	Required Materials / Filming Highlights
Phase	10 min	Venue Setup + Material Arrangement	Test projection, set up blank "Career Ecosystem Tree".	General Supervisor, Logistics	Sticky notes
Main Phase	20 min	Session 1: Ecosystem Roundtable Discussion	Team members role-play as experts from multiple fields to share insights.	Instructor, Assistant	Presentation PPT
Main Phase	20 min	Session 2: Drawing the Career Ecosystem Tree	Students connect their different strengths to the corresponding ecological niches.	Assistant	Sticky notes
Closing Phase	10 min	Summary Sharing	Results presentation, inspire diversity	Instructor	PPT

[Detailed Session Descriptions]

- **Session 1: Ecosystem Roundtable Discussion**
  - **Core Content:** The focus is not solely on the stories of scientists. Four roles are invited (either through live presentations, video recordings, or team members acting): the R&D Principal Investigator (PI), biosafety legal counsel, bioethicist, and technology entrepreneur. The narrative illustrates the various challenges encountered by a technology as it advances from plasmid design (the construction of artificial DNA molecules known as plasmids) to market launch.
  - **Key Expression:** "Synthetic biology (the interdisciplinary field of designing and constructing new biological parts, devices, and systems) offers more than just the path of becoming a PI. If you excel in critical thinking, you can serve as a bioethicist who defines the boundaries of technology. If you are knowledgeable about the market, you can be an entrepreneur who facilitates technology commercialization. Welcome to the complex ecosystem of the real world."
- **Session 2: Drawing the Career Ecosystem Tree**
  - **Core content:** Students are encouraged to freely select "skill tags" of different colors (e.g., communication skills, logical thinking, empathy, etc.). Guiding them to go to the front "Career Ecosystem Tree" poster, affixing their tags to the corresponding branches (career pathways).

- Key language: "The roots of this tree are synthetic biology, but its branches and leaves are vibrant and multicolored. Please find the position where your talents can shine."

VI、 Staffing and Division of Duties

- **General Supervisor, 1 person:** On-site control and time management
- **Five lecturers:** Responsible for explaining different "professional roles".
- **One teaching assistant:** Guides students in the collaborative creation of an ecological tree sticker activity.
- **Two persons assigned for photography and logistics:** Arrangement of backup supplies and photographing during the event.

VII、 List of Materials

- Publicity and Materials:
  - Small posters
- Equipment:
  - Laptop computers, projectors
  - Camera / High-definition mobile phone
- Teaching Materials:
  - Popular Science PPT
  - Sticky notes

VIII、 Risk Contingency Plans

- Equipment risks: Malfunction of projector or audio devices, or interruptions in normal operation. Equipment should be tested and adjusted in advance; backup equipment must be available. If the equipment cannot be repaired, the event will be simplified to the "lecturer's explanation plus distribution of materials" to ensure the core knowledge is delivered.
  Content delivery
- Venue risks: In the event that the original venue becomes unavailable, alternate venues should be activated, with prior confirmation from the alternative venue.

IX、 Effectiveness Evaluation

- Process evaluation: On-site observation of student participation and the level of interaction.
- Satisfaction evaluation: Collection of feedback forms completed by staff and random verbal inquiry of students' feedback.